

Latency Report

Pipeline: raw STT -> LLM -> TTS | Mode: LLM | Streaming: off | Sample size: 9 interactions

This measures the current non-streaming voice pipeline. Audio is recorded in the browser, sent to the backend for transcription, passed through the agent, synthesized to speech, and returned to the browser for playback.

Executive findings

- This is the raw baseline. Tool-backed turns can require one model pass to choose a tool and another model pass to verbalize the tool result.
- Local tool execution is effectively free in this demo because the tools call in-process sample data, not a remote database.
- The largest visible spike in this clean run is in the model layer: LLM total reaches 8.51s at p95.
- The CSV did not contain assistant text, so LLM replies in the cleaned table are reconstructed for review.

Summary metrics

Metric	What it means	Median	P95
End-of-turn detection	Delay after the user stops speaking before the turn is accepted.	0.68s	0.69s
Speech-to-text	Time to transcribe the captured user audio.	1.25s	2.85s
LLM first token	Time until the model starts producing output.	1.09s	1.65s
LLM total	Total model time. Tool turns can include planning plus post-tool response.	2.53s	8.51s
Local tool chain	Time spent executing local backend tools and mock lookups.	0.00s	0.01s
TTS first audio	Time until the backend receives first audio bytes from TTS.	1.23s	1.71s
TTS complete	Time until the full speech audio is generated.	2.14s	2.93s
Browser delivery	Client-side delivery and playback-start overhead after backend processing.	0.04s	0.45s
End-to-heard	Total delay from user finishing speaking to browser playback start.	6.61s	13.95s

No streaming (LLM mode) - cleaned interaction table

All timings below are shown in seconds. A dash means that stage did not apply to the turn.

#	User said	Agent replied	Tool?	End turn	STT	LLM	Tool chain	TTS first	TTS full	Heard
1	Hello	Hi there! How can I assist you today?	No	0.65s	2.85s	1.20s	-	1.23s	1.46s	6.61s
2	What can you do for me?	I can help with order status, ticket status, and arranging a support callback.	No	0.69s	1.04s	1.00s	-	0.97s	2.18s	4.96s
3	I need help with my order status.	May I have your full name so I can look up your order?	Yes	0.69s	1.25s	3.17s	0.00s	1.29s	1.76s	6.92s
4	My name is Arjun Mehta.	Thanks, Arjun. May I have your order ID?	Yes	0.69s	1.78s	2.70s	0.01s	1.00s	2.93s	8.16s
5	The order ID	I did not catch the full order ID. Please say the digits one at a time.	Yes	0.67s	0.95s	2.53s	0.00s	0.91s	1.45s	5.63s
6	The order ID is 2205.	I could not find that order for your account. I can connect you with a representative or schedu...	Yes	0.65s	1.41s	3.00s	0.00s	1.52s	2.39s	7.48s
7	Can you schedule a callback for me?	Sure, what day and time works best for the callback?	Yes	0.65s	1.00s	2.52s	0.00s	1.07s	1.90s	6.11s
8	Please schedule it for tomorrow after 7 PM.	Your callback is scheduled for October 1, 2026 at 7:30 PM.	Yes	0.68s	2.22s	8.51s	0.01s	1.52s	2.49s	13.95s
9	Okay, thank you.	You're welcome. Have a great day.	No	0.69s	1.22s	1.72s	-	1.71s	2.14s	5.80s

Latency Report

Pipeline: raw STT -> LLM -> TTS | Mode: speech directive | Streaming: off | Sample size: 9 interactions

This measures the current non-streaming voice pipeline. Audio is recorded in the browser, sent to the backend for transcription, passed through the agent, synthesized to speech, and returned to the browser for playback.

Executive findings

- Speech-directive mode keeps the same STT and local tool chain, but removes the post-tool LLM response pass for tool-backed turns.
- Tool-turn model latency drops from 2.85s median / 8.51s p95 in LLM mode to 1.84s median / 1.89s p95 in directive mode.
- TTS is similar in the clean comparison, so the measured improvement mainly comes from the model layer and deterministic tool responses.
- Directive tool replies are deterministic backend strings; non-tool greetings are reconstructed from the call flow.

Summary metrics

Metric	What it means	Median	P95
End-of-turn detection	Delay after the user stops speaking before the turn is accepted.	0.68s	0.69s
Speech-to-text	Time to transcribe the captured user audio.	1.37s	4.35s
LLM first token	Time until the model starts producing output.	1.46s	2.32s
LLM total	Total model time. Tool turns can include planning plus post-tool response.	1.77s	2.67s
Local tool chain	Time spent executing local backend tools and mock lookups.	0.00s	0.32s
TTS first audio	Time until the backend receives first audio bytes from TTS.	1.30s	2.54s
TTS complete	Time until the full speech audio is generated.	1.85s	3.07s
Browser delivery	Client-side delivery and playback-start overhead after backend processing.	0.04s	0.12s
End-to-heard	Total delay from user finishing speaking to browser playback start.	6.19s	8.76s

No streaming (directive mode) - cleaned interaction table

All timings below are shown in seconds. A dash means that stage did not apply to the turn.

#	User said	Agent replied	Tool?	End turn	STT	LLM	Tool chain	TTS first	TTS full	Heard
1	Hello.	Hi there! How can I assist you today?	No	0.65s	4.35s	1.77s	-	1.62s	1.85s	8.76s
2	What can you do for me?	I can help with order status, ticket status, and arranging a support callback.	No	0.68s	1.82s	1.46s	-	1.44s	2.15s	6.15s
3	I needed assistance with an update on my order.	I can help with that. May I have your full name?	No	0.65s	1.38s	2.67s	-	2.54s	2.89s	7.64s
4	Hello.	Hi there! How can I assist you today?	No	0.69s	2.48s	1.56s	-	0.81s	1.71s	6.47s
5	My name is Arjun Mehta.	May I have your order ID?	Yes	0.68s	1.31s	1.84s	0.32s	0.88s	1.52s	5.72s
6	The order ID is 2205.	I couldn't find order 2, 2, 0, 5 for this account. Would you like me to set up a callback?	Yes	0.68s	1.37s	1.89s	0.00s	1.50s	2.21s	6.19s
7	Yes please.	What day and time would work best for a callback?	Yes	0.68s	1.16s	1.84s	0.00s	1.30s	3.07s	6.78s
8	Tomorrow after 6 pm.	Your callback is scheduled for October 1, 2026 at 6:30 PM.	Yes	0.68s	1.04s	1.25s	0.01s	1.29s	1.77s	4.78s
9	Okay, thank you.	You're welcome. Have a great day.	No	0.69s	1.21s	0.90s	-	0.89s	1.41s	4.29s